

Akshayaa D

AI Engineer · Production RAG, Agentic AI & LLM Evaluation
+91 6374737790 | akshayaa.draj@gmail.com | LinkedIn | GitHub | Portfolio | Coimbatore, India

PROFESSIONAL SUMMARY

3+ years of engineering experience, currently an AI Engineer at FourKites, owning production RAG, LLM, and Agentic AI systems for enterprise supply-chain workflows end-to-end – turning ambiguous, high-stakes problems into measurable scale, reliability, and business-value gains.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Object-Oriented Programming

LLM & GenAI: Large Language Models (LLM), Transformers, Tokenization, Prompt Engineering, Tool/Function Calling, OpenAI API, Azure OpenAI (GPT-4o), GPT-4 Vision, Fine-Tuning Concepts, KV Cache

RAG & Vector Search: Document Ingestion, Chunking, Hybrid Search (BM25 + Semantic + Reciprocal Rank Fusion), Reranking, Embeddings, Self-Healing RAG, Qdrant, CLIP (Multimodal Embeddings)

Agentic Systems: LangChain, LangGraph, Multi-Agent Systems, Agentic Workflows, MCP (Model Context Protocol)

Automation & Tooling: Browser/Web Automation (Playwright), Pipedream, Claude Code

Backend & Cloud: FastAPI, Async Processing, Distributed Systems, Kafka, Redis, PostgreSQL, ClickHouse, Docker, Kubernetes, AWS/Azure, Azure Blob Storage, Git

Evaluation & Observability: Hybrid Retrieval + LLM-as-Judge Scoring, Agreement-Rate Benchmarking, Human-in-the-Loop, Observability, SigNoz, Chronosphere, Langfuse, LangSmith

PROFESSIONAL EXPERIENCE

AI Engineer II – FourKites, Chennai

June 2023 – Present

- Led end-to-end ownership of production agentic AI systems—**SAM** (multimodal RAG on LangChain/FastAPI/Kubernetes, CLIP + Qdrant retrieval, **1.39M+ executions**) and **ALAN** (AI booking agent on Pipedream/Playwright/Azure OpenAI vision).
- Replaced **SAM**'s hardcoded pipeline with a format-agnostic system (**21+ templates**, two-stage Azure OpenAI pipeline), fixing a multi-sheet bug that dropped **90%+** of file data and **28,693** garbage records while scaling **2.3x** at **0.1–0.3% failure**.
- Root-caused a **3.8x** excess in duplicate shipment creation to a write-before-read idempotency race across Kafka redeliveries and shipped a targeted fix, achieving a **99.6% reduction**—classified as P1 Avoided.
- Owned **ALAN**'s reliability across retailer and 3PL scheduling portals, retiring a **428-entry** hardcoded site registry in favor of recorded templates served at runtime, turning portal onboarding from an engineering task into a recording session—and cutting the booking failure rate from **60.7% to 8.9%** against a validated **~\$400K annual value**.
- Processed ambiguous, multilingual requirements—including Spanish-language email intake across **10,000+ messages**—to onboard a new enterprise customer, delivering Phase 2 in **2 months**; recognized by CTOs on both sides.
- Designed **ALAN**'s failure-recovery layer: two-level retry (in-process exponential backoff with jitter and a non-retryable error class, plus a Redis-backed workflow queue), Redis-based session reuse of authenticated browser state with dual idle/lifetime expiry, and a **3-hourly** validation cron reconciling appointment state against the portal—catching bookings that reported success but were never actually confirmed.

PERSONAL PROJECTS

PayMind | LangGraph, FastAPI, Qdrant, Hybrid Search (BM25+RRF), Gemini, RBAC, LangSmith

- Built a tool-calling agent that selects the right API among 112 via role-filtered hybrid search (BM25 + dense embeddings, RRF), scaling to **1,012** tools at **100%** top-5 retrieval accuracy and **~20ms** per search.
- Designed a code-level validator enforcing RBAC, parameter grounding (blocking hallucinated IDs/amounts), and human-in-the-loop confirmation for write actions – validated by **236** tests, zero LLM dependency.

Resume Reality Check | FastAPI, RAG, RRF, Evaluation, Gemini, ONNX, BM25, Docker

- Built a hybrid-retrieval (BM25 + semantic + Reciprocal Rank Fusion) system with an LLM-as-judge scored against a fixed rubric, separating what a resume claims from what it actually proves with cited evidence.
- Found and fixed **13** real production issues – retrieval corpus pollution, non-determinism from LLM serving-layer routing, memory-constrained deployment failure – validated against a **17-pair** hand-labeled eval set across **3** different resumes.

Swarm Onboard | Python, FastAPI, LangGraph, MCP, Redis, Pydantic, LangSmith, AST Parsing, GitHub REST API

- Designed a multi-agent system that analyzes unfamiliar GitHub repositories and generates onboarding documentation, using a deterministic graph-building layer plus an LLM-based reasoning layer with cross-agent validation.
- Verified generated documentation against **4** different GitHub repositories varying in size and structure to confirm output quality and consistency.

EDUCATION

Sri Krishna College of Engineering and Technology

Bachelor of Technology (B.Tech), Information Technology CGPA: 9.34/10.0

Jul 2019 – May 2023

Coimbatore, India

5th Place, HackCovid 2021 Hackathon – built the website for a frontline-worker mental-health platform (team of 4).